

Problem 10.24

Calculate k if a deposit of 1 will accumulate to 2.7183 in 10 years at a force of interest given by

$$\delta_t = \begin{cases} kt, & 0 < t \leq 5, \\ 0.04kt^2, & t > 5. \end{cases}$$

Solution

For a time-varying force of interest, the accumulation factor over 10 years is

$$\exp \left\{ \int_0^{10} \delta_t dt \right\}.$$

Since the force changes at time 5, we split the integral:

$$\exp \left\{ \int_0^5 kt dt + \int_5^{10} 0.04kt^2 dt \right\} = 2.7183.$$

The first integral is

$$\int_0^5 kt dt = \left. \frac{kt^2}{2} \right|_0^5 = \frac{25k}{2}.$$

The second integral is

$$\begin{aligned} \int_5^{10} 0.04kt^2 dt &= \left. \frac{0.04kt^3}{3} \right|_5^{10} \\ &= \frac{0.04k}{3} (10^3 - 5^3) \\ &= \frac{0.04k}{3} (875) \\ &= \frac{35k}{3}. \end{aligned}$$

Therefore,

$$\exp \left\{ \frac{25k}{2} + \frac{35k}{3} \right\} = 2.7183.$$

Taking natural logarithms gives

$$\begin{aligned} \frac{25k}{2} + \frac{35k}{3} &= \ln(2.7183), \\ \frac{145k}{6} &= \ln(2.7183). \end{aligned}$$

Hence,

$$\begin{aligned} k &= \frac{6 \ln(2.7183)}{145} \\ &\approx 0.04137958696. \end{aligned}$$

Thus,

$$\boxed{k \approx 0.041379587}.$$